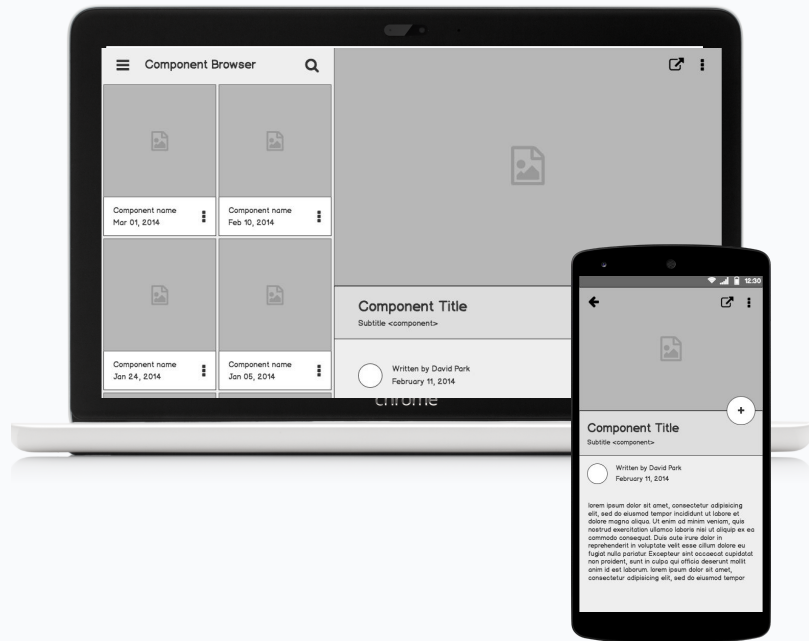


Multi-Brand Marketing Performance Analysis & Prediction

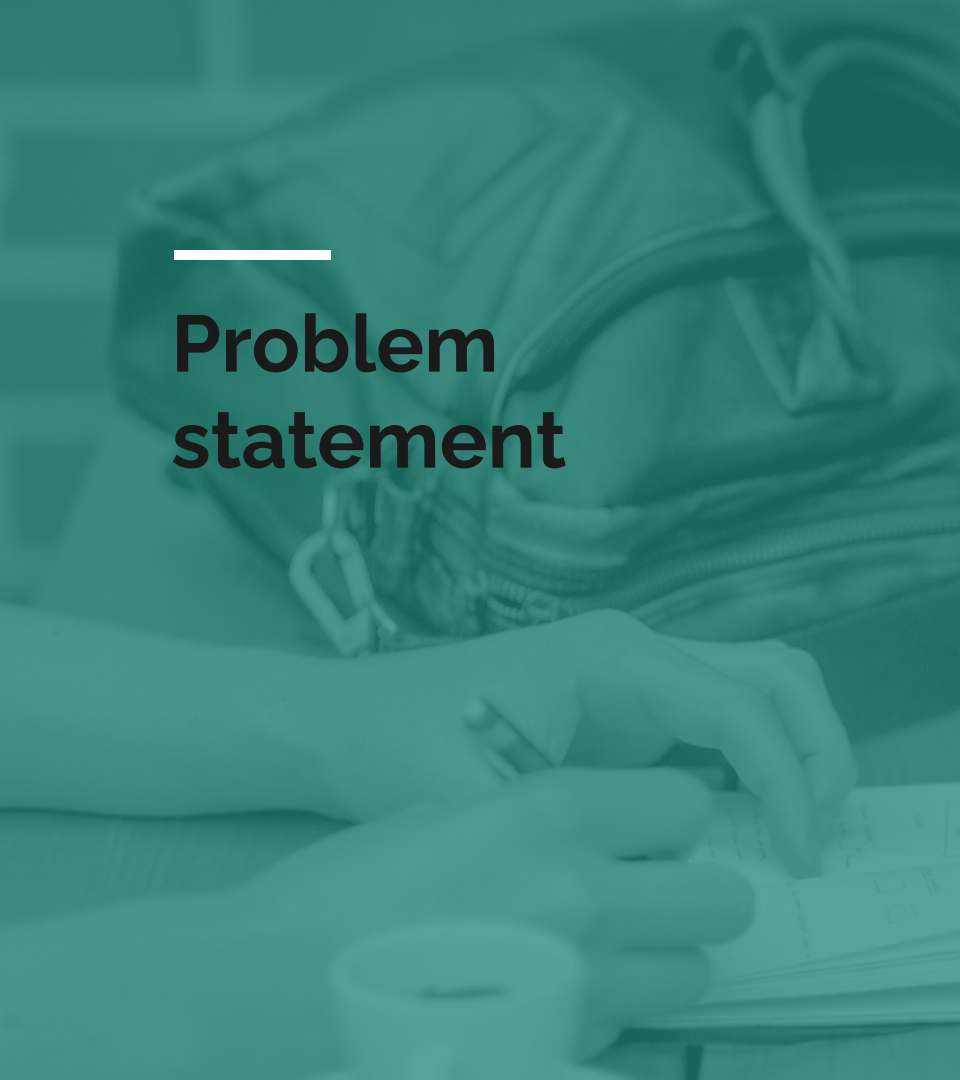
An End-to-End Machine Learning & Analytics Pipeline





Machine Learning: Predictive Pattern Modeling

Advanced Analytics for Data-Driven Intelligence

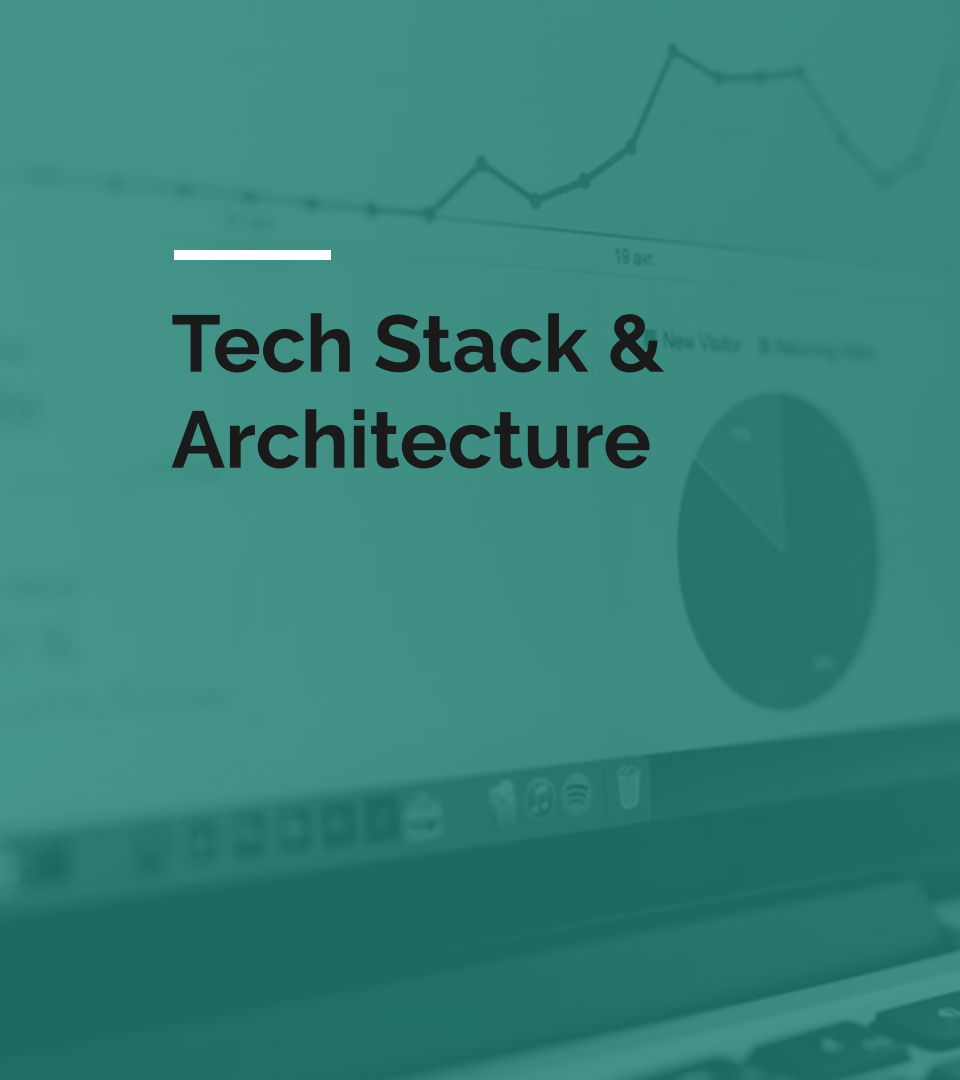


Problem statement

Nykaa, Purple, Tira generate massive campaign data volumes.

Raw CSV data is unstructured with missing values.

Automate pipeline to predict future campaign profitability insights.



Tech Stack & Architecture

Data Manipulation: Python, Pandas, NumPy

Database & Querying: PostgreSQL (pgAdmin)

Data Visualization: Matplotlib, Seaborn

Machine Learning: Scikit-Learn (Regression & Classification)

Deployment: Streamlit (Interactive Web Dashboard)

Data Preprocessing & Engineering

Cleaning and Structuring the Data



Data Cleaning

Removed duplicates and handled missing values, reducing dataset to ~75,500 records.



Mathematical Validation

Recalculated Return on Investment (ROI) to ensure financial accuracy.



Feature Engineering

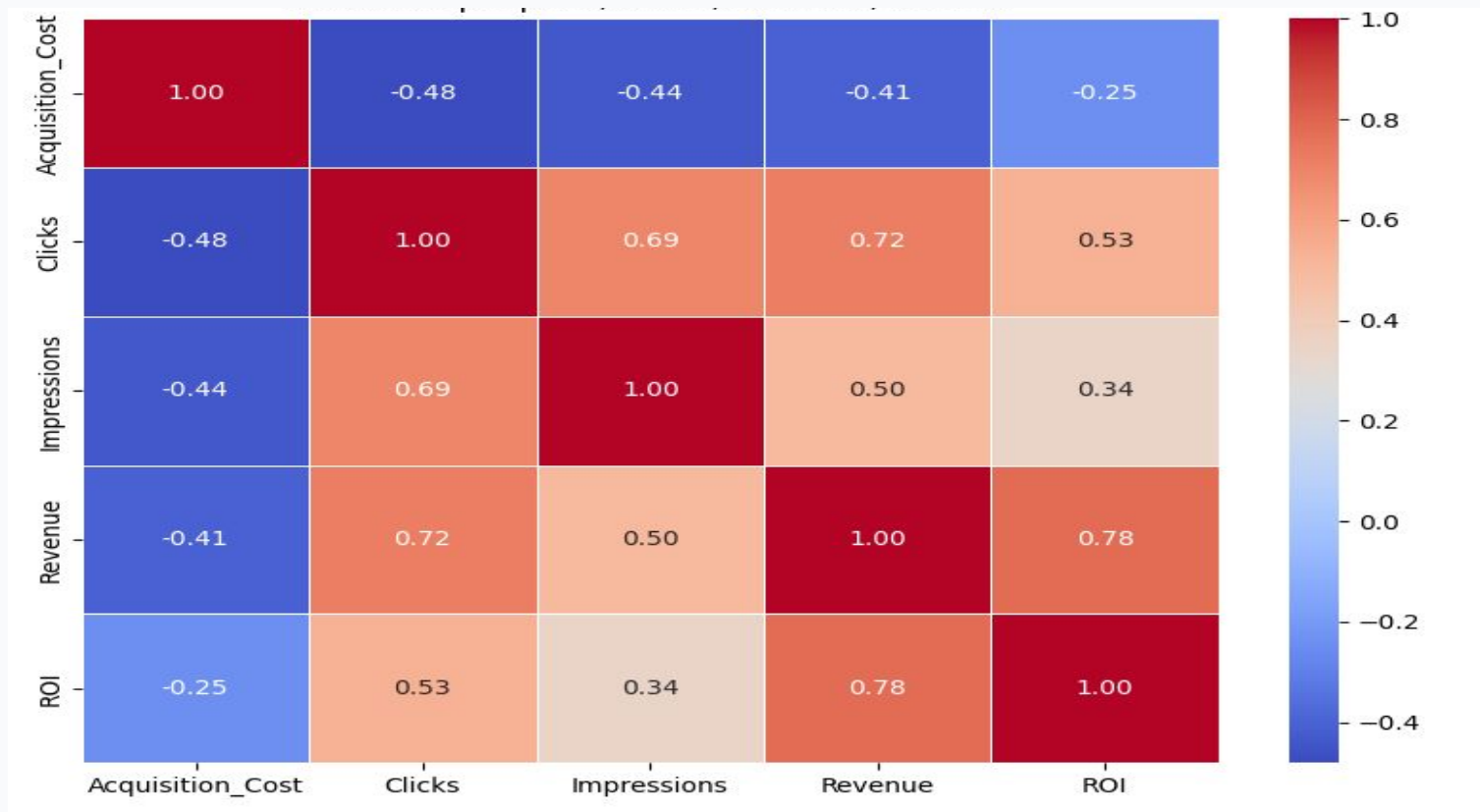
Engineered binary **Profit_Flag** (1 for Profit, 0 for Loss) based on ROI.



Multi-Label Encoding

Transformed **Channel_Used** into machine-readable binary columns.

Exploratory Data Analysis & SQL Insights



Exploratory Data Analysis & SQL Insights

Data Output Messages Notifications



SQL

Showing rows: 1 to 5



	campaign_id character varying (50) 🔒	target_audience character varying (50) 🔒	revenue double precision 🔒	acquisition_cost double precision 🔒	roi double precision 🔒
1	TI-CMP-8939	Youth	8695	6717.78	-0.97
2	TI-CMP-16012	Youth	9625	8183.09	-0.97
3	NY-CMP-24406	Premium Shoppers	6183	6805.56	-0.97
4	TI-CMP-1359	Working Women	9180	8310.08	-0.97
5	NY-CMP-53545	College Students	9386	6675.92	-0.96

Predictive Modeling (Machine Learning)

Building the AI Models



Revenue Prediction

Regression Analysis

Algorithm: Linear Regression

Performance: Achieved an R^2 score of 0.77.



Profitability Prediction

Classification Analysis

Algorithm: Logistic Regression

Performance: 91.1% Accuracy and an F1-Score of 0.94.

Note: Explicitly dropped the ROI column during training to prevent data leakage and ensure model integrity.

Deployment & Business Impact

Campaign Settings

Duration (Days)
10

Expected Impressions
500

Expected Clicks
50

Expected Leads
20

Expected Conversions
10

Acquisition Cost (Spend)
350

Target Engagement Score
20.33



Prediction Results

Predicted Revenue

₹ 11,896.17

Profitable Campaign (Profit)

Key Input Metrics Overview

Marketing Funnel Inputs

